

Chi square test for Part one: (Table 1 ; Knowledge)

Q.1. Q.1. Are you aware of the existence of genetic diagnosis in your family? / or before having an affected child (incase for group 4)At which stage did you have this information?

- **Need to know the statistical significant in the awarnass (who answer aware) for groups 1,2,3,4**

		MS marital status				Total
		Single or Engaged	Married\ divorced\ widower with no kids	Married\ divorced\ widower with healthy kids	Married\ divorced\ widower with affected kids	
هل أنت على علم بوجود Q4	لا	11	7	49	225	292
تشخيص بمرض جيني وراثي لديكم في العائلة؟	نعم	115	45	181	26	367
Total		126	52	230	251	659

Chi-Square Tests

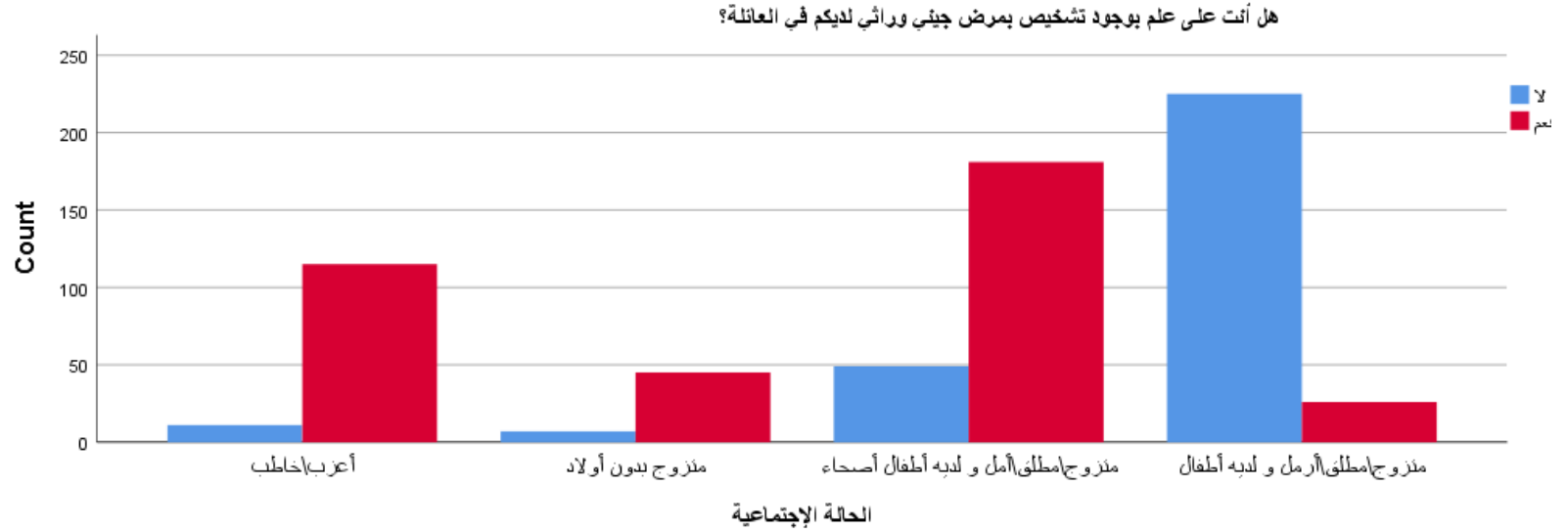
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	343.048 ^a	3	.000
Likelihood Ratio	383.899	3	.000
Linear-by-Linear Association	242.826	1	.000
N of Valid Cases	659		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 23.04.

Symmetric Measures

		Value	Approximate Significance
Nominal by Nominal	Phi	.721	.000
	Cramer's V	.721	.000
	Contingency Coefficient	.585	.000
N of Valid Cases		659	

From the chi-square test, there is a significant correlation between the marital status and the awareness of the existence of genetic diagnosis in the family? / or before having an affected child at 95% confidence level. According to the Phi and Cramer's V test this correlation appears to be strong, where Phi and Cramer's V are positively defined measurement that lies between 0 and 1 to measure the strength of the correlation of 2 variables in a contingency tables where 1 means that the variables are perfectly correlated and 0 means that they are independent.



As presented in the previous graph, people who are married and have affected children are the most group that suffer from the non-awareness of the existence of genetic diagnosis in their families. On the other hand, people who are married and have healthy kids are the most group that have awareness of the existence of genetic diagnosis in their families.

This highlights the importance of increasing the awareness of genetic diagnosis for couples who are about to get married.

- **Between groups 2 ,3,4 need to know the statistical significance in the awarnass before marriage and after marriage**
Group 2:

	divorced with no kids	married with no kids	
لا	0	7	7

هل أنت على علم بوجود Q4	نعم	1	44	45
تشخيص بمرض جيني وراثي				
لديكم في العائلة؟				
Total		1	51	52

	Value	df	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	.159 ^a	1	.690		
Continuity Correction ^b	.000	1	1.000		
Likelihood Ratio	.292	1	.589		
Fisher's Exact Test				1.000	.865
Linear-by-Linear Association	.156	1	.693		
N of Valid Cases	52				

a. 2 cells (50.0%) have expected count less than 5. The minimum expected count is .13.

b. Computed only for a 2x2 table

In this case, the Fisher exact test was used instead of the Chi-Square test because there was more than 20% of the expected counts that were less than 5. From the Fisher Exact test, the awareness of the existence of genetic diagnosis in the family is totally independent from whether the couple are married or not i.e., divorced or widowed in the “married with no kids” group.

Group 3:

Marital_status	حالتك الاجتماعية:	
divorced\		
widower with	married with	
healthy kids	healthy kids	Total

هل أنت على علم بوجود Q4	لا	0	49	49
تشخيص بمرض جيني وراثي لديكم في العائلة؟	نعم	2	179	181
Total		2	228	230

Chi-Square Tests

	Value	df	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	.546 ^a	1	.460		
Continuity Correction ^b	.000	1	1.000		
Likelihood Ratio	.963	1	.326		
Fisher's Exact Test				1.000	.619
Linear-by-Linear Association	.544	1	.461		
N of Valid Cases	230				

a. 2 cells (50.0%) have expected count less than 5. The minimum expected count is .43.

b. Computed only for a 2x2 table

In this case, the Fisher exact test was used instead of the Chi-Square test because there was more than 20% of the expected counts that were less than 5. From the Fisher Exact test, the awareness of the existence of genetic diagnosis in the family is totally independent from whether the couple are married or not i.e., divorced or widowed in the “married with healthy kids” group.

Group 4:

Marital_status : حالتك الاجتماعية | Total

		divorced\ widower with affected kids	married with affected kids	
هل أنت على علم بوجود Q4	لا	5	220	225
تشخيص بمرض جيني وراثي لديكم في العائلة؟	نعم	1	25	26
Total		6	245	251

Chi-Square Tests

	Value	df	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	.263 ^a	1	.608		
Continuity Correction ^b	.000	1	1.000		
Likelihood Ratio	.228	1	.633		
Fisher's Exact Test				.485	.485
Linear-by-Linear Association	.262	1	.608		
N of Valid Cases	251				

a. 1 cells (25.0%) have expected count less than 5. The minimum expected count is .62.

b. Computed only for a 2x2 table

In this case, the Fisher exact test was used instead of the Chi-Square test because there was more than 20% of the expected counts that were less than 5. From the Fisher Exact test, the awareness of the existence of genetic diagnosis in the family is totally independent from whether the couple are married or not i.e., divorced or widowed in the “married with affected kids” group.

- **Need to know the statistical significant in non awareness (who answer not aware) 1,2,3,4**

Q.2.Are you aware of the existence of premarital genetic testing for diseases diagnosed in your family? And, at which stage, did you have this information?

- **Need to know the statistical significant in the awarnass (who answer aware) for groups 1,2,3,4**

هل أنت على دراية بوجود خدمة الفحص الجيني قبل الزواج للمرض المشخص لديكم في Q7
MS marital status Crosstabulation * العائلة ؟

Count

		MS marital status				
		Single or Engaged	Married\ divorced\ widower with no kids	Married\ divorced\ widower with healthy kids	Married\ divorced\ widower with affected kids	Total
Q7	لا هل أنت على دراية بوجود	15	9	88	188	300
	نعم خدمة الفحص الجيني قبل الزواج للمرض المشخص لديكم في العائلة ؟	111	36	142	63	352
Total		126	45	230	251	652

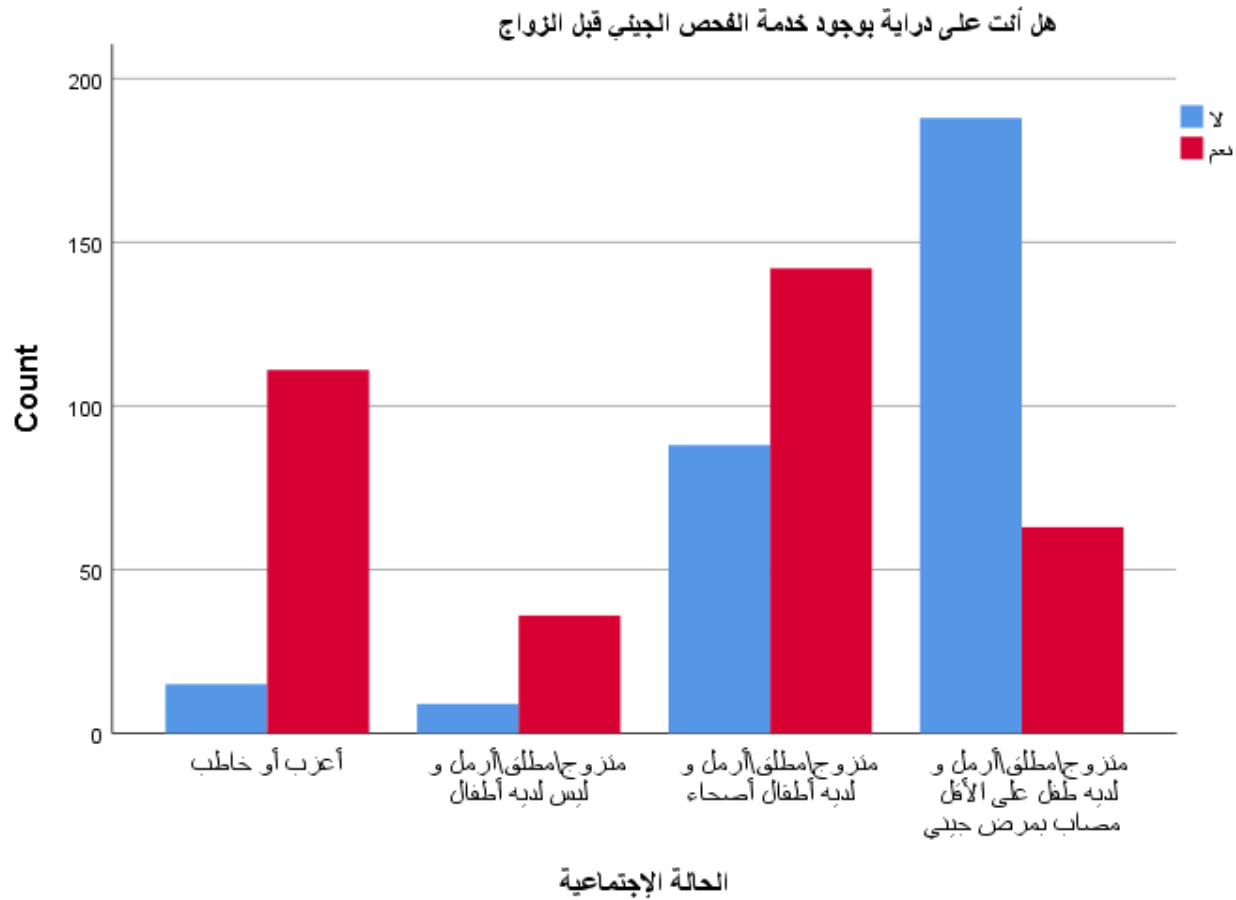
Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	161.150 ^a	3	.000
Likelihood Ratio	173.800	3	.000
Linear-by-Linear Association	146.078	1	.000
N of Valid Cases	652		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 20.71.

Symmetric Measures		Value	Approximate Significance
Nominal by Nominal	Phi	.497	.000
	Cramer's V	.497	.000
	Contingency Coefficient	.445	.000
N of Valid Cases		652	

From the chi-square test, there is a significant correlation between the marital status and the awareness of the existence of premarital genetic testing for diseases diagnosed in the family at 95% confidence level. According to the Phi and Cramer's V test this correlation appears to be moderate.



As presented in the previous graph, people who are married and have affected children are the most group that suffer from the non-awareness of the existence of premarital genetic testing for diseases diagnosed in their families. On the other hand, people who are married and have healthy kids are the most group that have awareness the existence of premarital genetic testing for diseases diagnosed in their families.

This highlights the importance of increasing the awareness of premarital genetic testing for couples who are about to get married.

- **For groups 2 and 3 need need to know the statistical significance in the awarnass before marriage and after marriage.**

Group 2:

هل أنت على دراية بوجود خدمة الفحص الجيني قبل الزواج Q7
حالتك * Marital_status * للمرض المشخص لديكم في العائلة؟
الاجتماعية: Crosstabulation

Count

		Marital_status : حالتك الاجتماعية		Total
		devorced with no kids	married with no kids	
Q7 هل أنت على دراية بوجود	لا	0	9	9
خدمة الفحص الجيني قبل الزواج للمرض المشخص لديكم في العائلة؟	نعم	1	35	36
Total		1	44	45

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	.256 ^a	1	.613		

Continuity Correction ^b	.000	1	1.000		
Likelihood Ratio	.452	1	.501		
Fisher's Exact Test				1.000	.800
Linear-by-Linear Association	.250	1	.617		
N of Valid Cases	45				

a. 2 cells (50.0%) have expected count less than 5. The minimum expected count is .20.

b. Computed only for a 2x2 table

From the Fisher Exact test, the awareness of premarital genetic testing in the family is totally independent from whether the couple are married or not i.e., divorced or widowed in the “married with no kids” group.

Group 3:

هل أنت على دراية بوجود خدمة الفحص الجيني قبل الزواج Q7
حالتك Marital_status * للمرض المشخص لديكم في العائلة؟
الاجتماعية: Crosstabulation

Count

		حالتك الاجتماعية Marital_status		Total
		divorced\widower with healthy kids	married with healthy kids	
لا	هل أنت على دراية بوجود Q7	0	88	88
نعم	خدمة الفحص الجيني قبل الزواج للمرض المشخص لديكم في العائلة؟	2	140	142
Total		2	228	230

Chi-Square Tests					
	Value	df	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	1.250 ^a	1	.263		
Continuity Correction ^b	.150	1	.698		
Likelihood Ratio	1.940	1	.164		
Fisher's Exact Test				.525	.380
Linear-by-Linear Association	1.245	1	.265		
N of Valid Cases	230				

a. 2 cells (50.0%) have expected count less than 5. The minimum expected count is .77.

b. Computed only for a 2x2 table

From the Fisher Exact test, the awareness of premarital genetic testing in the family is totally independent from whether the couple are married or not i.e., divorced or widowed in the “married with healthy kids” group.

- **Need to know the statistical significant in non awareness (who answer not aware) 1,2,3,4**

Chi square test for Part two : (Table 2 ; Attitude and practice)

For statement “Premarital genetic testing could act as a preventative option in the families with rare genetic disease” chooses the option that describes your believe about the statement

- **Need to know the statistical significant in the answers between groups 1,2,3,4**

Total

		Single or Engaged	Married\ divorced\ widower with no kids	Married\ divorced\ widower healthy kids	
Q11 إلى أي مدى توافق على	أوافق	16	5	22	43
الرأي القائل : " يعد الفحص	أوافق بشدة	104	47	205	356
الجيني قبل الزواج إجراءا وقائيا	لا أوافق	1	0	0	1
لتجنب تكرار الإصابة بالمرض	لا أوافق	1	0	0	1
الوراثي للأجيال مستقبلا في	بشدة				
العوائل المشخص لديها أفراد	محايد	4	0	3	7
بمرض جيني وراثي " ؟					
Total		126	52	230	408

Chi-Square Tests

				Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
			Asymptotic		99% Confidence			99% Confidence Interval	
			Significance		Lower	Upper			
	Value	df	(2-sided)	Significance	Bound	Bound	Significance	Lower Bound	Upper Bound
Pearson Chi-Square	8.437 ^a	8	.392	.405 ^b	.392	.418			
Likelihood Ratio	9.235	8	.323	.319 ^b	.307	.331			
Fisher's Exact Test	8.237			.389 ^b	.377	.402			
Linear-by-Linear Association	.575 ^c	1	.448	.464 ^b	.451	.477	.241 ^b	.230	.252
N of Valid Cases	408								

a. 9 cells (60.0%) have expected count less than 5. The minimum expected count is .13.

b. Based on 10000 sampled tables with starting seed 957002199.

c. The standardized statistic is -.758.

In this case, non of the Chi-square or Fisher exact test can be used. This is because 60.0% of the expected count are less than 5. In this case 10,000 Monte Carlo samples were used inferred by the original sample. The Chi-square of these samples showed that there is no significant correlation between the agreement on the statement “Prenatal genetic testing could act as a preventative option in the families with rare genetic disease” and the marital status of the 3 **groups.**

For statement “The history of the genetic disease in your family affected your chances and choices for marriage” select the option that describes your believe a bout the statement choices for the marriage.

- Need to know the statistical significance in the answers between groups 1,2,3, and 4.
-

وجود التشخيص بالمرض الجيني لديكم في العائلة قد أثر على فرصك وخياراتك " Q27
MS Marital_status Crosstabulation * المتعلقة بالزواج

Count

		MS Marital_status			Total
		Single or Engaged	Married\ divorced\ widower with no kids	Married\ divorced\ widower healthy kids	
وجود التشخيص بالمرض " Q27	أوافق	38	10	55	103
	أوافق بشدة	26	14	40	80
	على فرصك وخياراتك المتعلقة	0	0	57	57
	" بالزواج	0	0	18	18
	غير موافق	0	0	18	18
	بشدة	0	0	18	18
	لا أوافق	24	7	0	31
	لا أوافق بشدة	5	3	0	8

محايد	33	15	60	108
Total	126	49	230	405

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)		Monte Carlo Sig. (1-sided)			
				Significance	99% Confidence Interval		Significance	99% Confidence Interval	
					Lower Bound	Upper Bound		Lower Bound	Upper Bound
Pearson Chi-Square	114.952 ^a	12	.000	.000 ^b	.000	.000			
Likelihood Ratio	155.641	12	.000	.000 ^b	.000	.000			
Fisher's Exact Test	137.841			.000 ^b	.000	.000			
Linear-by-Linear Association	1.303 ^c	1	.254	.259 ^b	.247	.270	.130 ^b	.121	.138
N of Valid Cases	405								

- a. 5 cells (23.8%) have expected count less than 5. The minimum expected count is .97.
- b. Based on 10000 sampled tables with starting seed 1507486128.
- c. The standardized statistic is -1.141.

Same to the previous case, non of the Chi-square or Fisher exact test can be used and again the Monte Carlo samples were used. The Chi-square of these samples showed that there is a significant correlation between the agreement on the statement “The history of the genetic disease in your family affected your chances and choices for marriage” and the marital status of the **3 groups**.

Symmetric Measures

Value	Asymptotic Standard Error ^a	Approximate T ^b	Approximate Significance	Monte Carlo Significance	Monte Carlo Significance 99% Confidence Interval
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							Lower Bound	Upper Bound
Nominal by Nominal	Phi	.533			.000	.000 ^c	.000	.000
	Cramer's V	.377			.000	.000 ^c	.000	.000
	Contingency Coefficient	.470			.000	.000 ^c	.000	.000
Interval by Interval	Pearson's R	-.057	.050	-1.142	.254 ^d	.259 ^c	.247	.270
Ordinal by Ordinal	Spearman Correlation	-.008	.052	-.165	.869 ^d	.872 ^c	.863	.881
N of Valid Cases		405						

- a. Not assuming the null hypothesis.
- b. Using the asymptotic standard error assuming the null hypothesis.
- c. Based on 10000 sampled tables with starting seed 1507486128.
- d. Based on normal approximation.

The correlation strength is then measured by the Phi and Cramer's V correlation measurements using the Monte Carlo samples. However both of the measurements showed moderate correlation between the agreement on the statement "The history of the genetic disease in your family affected your chances and choices for marriage" and the marital status of the **3 groups**.

For statement What do you think is the appropriate timing for premarital genetic testing?

- Need to know the statistical significance in the answers between groups 1,2,3, and 4.
-

متى باعتقادك أنسب وقت أو مرحلة لإجراء الفحص الجيني لمعرفة إمكانية أن يكون الشخص حاملاً أو غير Q28
 حامل للتغير الجيني المسبب للمرض؟ * MS Marital_status Crosstabulation

Count

		MS Marital_status				
		Single or Engaged	Married\divorced\wedower with no kids	Married\divorced\wedower healthy kids	Married\divorced\wedower with affected kids	Total
متى باعتقادك أنسب وقت Q28	بعد إتمام الزواج	1	0	0	2	3
أو مرحلة لإجراء الفحص الجيني	في عمر 18 سنة	0	26	116	90	232
لمعرفة إمكانية أن يكون الشخص	في عمر 18 سنه	73	0	0	0	73
حاملًا أو غير حامل للتغير الجيني	قبل البدء بإجراءات الزواج	45	25	0	0	70
المسبب للمرض ؟	قبل البدء بإجراءات الزواج	0	1	102	143	246
	لا أعلم متى الوقت المناسب	7	0	12	16	35
Total		126	52	230	251	659

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
				Significance	99% Confidence Interval		Significance	99% Confidence Interval	
					Lower Bound	Upper Bound		Lower Bound	Upper Bound
Pearson Chi-Square	665.065 ^a	15	.000	.000 ^b	.000	.000			
Likelihood Ratio	690.650	15	.000	.000 ^b	.000	.000			
Fisher's Exact Test	654.682			.000 ^b	.000	.000			
Linear-by-Linear Association	13.011 ^c	1	.000	.000 ^b	.000	.001	.000 ^b	.000	.000
N of Valid Cases	659								

a. 5 cells (20.8%) have expected count less than 5. The minimum expected count is .24.

b. Based on 10000 sampled tables with starting seed 1122541128.

c. The standardized statistic is 3.607.

Same to the previous case, non of the Chi-square or Fisher exact test can be used and again the Monte Carlo samples were used. The Chi-square of these samples showed that there is a significant correlation between the participants opinions about what do they think is the appropriate timing for premarital genetic testing and the marital status.

		Symmetric Measures						
		Value	Asymptotic Standard Error ^a	Approximate T ^b	Approximate Significance	Monte Carlo Significance 99% Confidence Interval	Lower Bound	Upper Bound
Nominal by Nominal	Phi	1.005			.000	.000 ^c	.000	.000
	Cramer's V	.580			.000	.000 ^c	.000	.000
	Contingency Coefficient	.709			.000	.000 ^c	.000	.000
Interval by Interval	Pearson's R	.141	.031	3.641	.000 ^d	.000 ^c	.000	.001
Ordinal by Ordinal	Spearman Correlation	.163	.038	4.235	.000 ^d	.000 ^c	.000	.000
N of Valid Cases		659						

a. Not assuming the null hypothesis.

b. Using the asymptotic standard error assuming the null hypothesis.

c. Based on 10000 sampled tables with starting seed 1122541128.

d. Based on normal approximation.

The correlation strength is then measured by the Phi and Cramer's V correlation measurements using the Monte Carlo samples. However the measurements showed moderate to high correlation between the participants opinions about what do they think is the appropriate timing for premarital genetic testing and the marital status.

For statement What your advice to at-risk couple who are both carriers and at risk to have an affected child

- Need to know the statistical significance in the answers between groups 1,2,3, and 4.
-

بماذا تنصح أي شخصين مقبلين على الزواج وتمت معرفة _ عن طريق الفحص_ أن كلاهما حامل للتغير الجيني Q29
MS Marital_status * :المسبب للمرض وأن أطفالهما معرضين للإصابة بالمرض الجيني الذي سبق وأن شخص في الأسرة
Crosstabulation

Count

		MS Marital_status				
		Single or Engaged	Married\divorced\wedower with no kids	Married\divorced\wedower healthy kids	Married\divorced\wedower with affected kids	Total
بماذا تنصح أي شخصين Q29	أنصحهم بإتمام الزواج إن كان هناك رابط عاطفي بينهما	1	0	0	2	3
مقبلين على الزواج وتمت معرفة _ عن طريق الفحص_ أن كلاهما حامل للتغير الجيني المسبب للمرض وأن أطفالهما معرضين للإصابة بالمرض الجيني الذي سبق: وأن شخص في الأسرة:	أنصحهم بإتمام الزواج إن كان هناك ضغط من الاسرة	0	1	0	0	1
	أنصحهم بإتمام الزواج والتوكل على الله	3	3	1	3	10
	أنصحهم بإلغاء مشروع الخطبة أو الزواج	29	13	68	69	179
	أنصحهم بالتوجه لأخذ المشورة الوراثية من شخص متخصص بمجال الأمراض ا	0	30	0	0	30
	أنصحهم بالتوجه لأخذ المشورة الوراثية من متخصص بمجال الأمراض الوراثية	77	0	134	0	211
	أنصحهم بالتوجه لاخت المشورة الوراثية من شخص متخصص بالامراض الوراثية	0	0	0	157	157

1	0	0	1	0	اذا كان بينهم رابط عاطفي يبحثوا عن بدائل في الانجاب واذا كان ليس ه
1	0	1	0	0	الإتفاق بالزواج على اساس عدم الانجاب.
21	20	0	1	0	القرار يعتمد عل طبيعة المرض وشدة أعراضه في حالة إصابة الاطفال به
15	0	0	0	15	القرار يعتمد على طبيعة المرض وشدة أعراضه في حالة إصابة الاطفال به
21	0	21	0	0	القرار يعتمد على طبيعة المرض وشدة أعراضه في حالة إصابة الأطفال به
3	0	0	3	0	القرار يعتمد على طبيعة المرض وشدة أعراضه في حالة إصابة الاطفال به
1	0	1	0	0	انصحهم بالتوجه الى العيادة ومقابلة الشخص المختص واستيضاح الامور
1	0	1	0	0	اولا اخذ النصيحة من المرشدة الوراثية و بعد ذلك يعتمد ع طبيعة المرض
1	0	1	0	0	تجنب الزواج من الأقارب في حال حمل المرض الجيني
1	0	1	0	0	عدم الارتباط الا اذا كان هناك علاقة عاطفية قوية، فالزواج هو الحل ب

لا يستطيع ان انصحهم بأي شي لانهم ان كانوا مرتبطين عاطفيا فلن يستمع	1	0	0	0	1
وإذا كان هناك خطه علاجه تساعد في حل المرض وتقليل حدة الخطورة يفضل	0	0	1	0	1
Total	126	52	230	251	659

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
				Significance	Lower Bound	Upper Bound	Significance	Lower Bound	Upper Bound
Pearson Chi-Square	1003.603 ^a	54	.000	.000 ^b	.000	.000			
Likelihood Ratio	897.821	54	.000	.000 ^b	.000	.000			
Fisher's Exact Test	845.058			.000 ^b	.000	.000			
Linear-by-Linear Association	2.406 ^c	1	.121	.122 ^b	.113	.130	.060 ^b	.054	.066
N of Valid Cases	659								

a. 55 cells (72.4%) have expected count less than 5. The minimum expected count is .08.

b. Based on 10000 sampled tables with starting seed 605580418.

c. The standardized statistic is 1.551.

Same to the previous case, non of the Chi-square or Fisher exact test can be used and again the Monte Carlo samples were used. The Chi-square of these samples showed that there is a significant correlation between the participants advice to at-risk couple who are both carriers and at risk to have an affected child and the marital status.

		Symmetric Measures					
		Value	Asymptotic Standard Error ^a	Approximate T ^b	Approximate Significance	Monte Carlo Significance	99% Confidence Interval
						Significance	Lower Bound Upper Bound
Nominal by Nominal	Phi	1.234			.000	.000 ^c	.000 .000
	Cramer's V	.712			.000	.000 ^c	.000 .000
	Contingency Coefficient	.777			.000	.000 ^c	.000 .000
Interval by Interval	Pearson's R	.060	.038	1.553	.121 ^d	.122 ^c	.113 .130
Ordinal by Ordinal	Spearman Correlation	.263	.042	6.978	.000 ^d	.000 ^c	.000 .000
N of Valid Cases		659					

- a. Not assuming the null hypothesis.
- b. Using the asymptotic standard error assuming the null hypothesis.
- c. Based on 10000 sampled tables with starting seed 605580418.
- d. Based on normal approximation.

The correlation strength is then measured by the Phi and Cramer's V correlation measurements using the Monte Carlo samples. However the measurements showed high correlation between the participants advice to at-risk couple who are both carriers and at risk to have an affected child and the marital status.

For statement : "To which extend you agree with this statement "The premarital genetic testing for rare genetic disorders should be voluntary not mandatory"

- **Need to know the statistical significance in the answers between groups 1,2,3, and 4.**

**إلى أي مدى توافق على الرأي القائل: "يجب أن يكون الفحص الجيني قبل الزواج للمرض Q30
MS Marital_status Crosstabulation * الوراثة إجباريا وليس إجباريا**

Count

		MS Marital_status				
		Single or Engaged	Married\divorced\wedower with no kids	Married\divorced\wedower healthy kids	Married\divorced\wedower with affected kids	Total
إلى أي مدى توافق على Q30	أوافق	15	8	30	41	94
الرأي القائل:" يجب أن يكون الفحص الجيني قبل الزواج	أوافق بشدة	13	3	27	44	87
للمرض الوراثي إختياريا وليس إجباريا	لا أوافق	40	13	65	61	179
	لا أوافق بشدة	39	18	74	76	207
	محايد	19	10	34	29	92
Total		126	52	230	251	659

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	12.982 ^a	12	.370
Likelihood Ratio	13.239	12	.352
Linear-by-Linear Association	3.910	1	.048
N of Valid Cases	659		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 6.86.

From the chi-square test, there is no significant correlation between the marital status and agreement with this statement
 “The premarital genetic testing for rare genetic disorders should be voluntary not mandatory” at 95% confidence level..

Chi square test for part 3 (Comparison between carriers and Non carriers in groups 1,2 and 3)

- **For carriers :** I would like to know the statistical significance in the answers across all the four statements in table 3.
- A) It was easy to disclose carrier status result to the family

Group1 * Agreement1 Crosstabulation

Count

		Agreement1					
		Agree	Disagree	Neutral	Strongly Agree	Strongly disagree	Total
Group1	Group 1	10	2	4	22	1	39
	Group 2	5	3	3	9	0	20
	Group 3	20	3	11	27	0	61
Total		35	8	18	58	1	120

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)		Monte Carlo Sig. (1-sided)		
				Significance	99% Confidence Interval	Significance	99% Confidence Interval	
					Lower Bound	Upper Bound	Lower Bound	Upper Bound
Pearson Chi-Square	6.878 ^a	8	.550	.581 ^b	.569	.594		
Likelihood Ratio	6.572	8	.583	.667 ^b	.655	.679		
Fisher's Exact Test	6.711			.588 ^b	.576	.601		

Linear-by-Linear Association	1.293 ^c	1	.256	.256 ^b	.245	.267	.132 ^b	.124	.141
N of Valid Cases	120								

- a. 7 cells (46.7%) have expected count less than 5. The minimum expected count is .17.
- b. Based on 10000 sampled tables with starting seed 1615198575.
- c. The standardized statistic is -1.137.

The Chi-square of these Mone Carlo samples showed at 95% confidence level that there is no significant correlation between the agreement degree on if it was easy to disclose carrier status result to the family and the marital status.

B) It was easy to disclose carrier status result to the spouse, or it will be easy to disclose result to you're a future spouse

Group1 * Agreement1 Crosstabulation

Count

		Agreement1				
		Agree	Neutral	Strongly Agree	Strongly disagree	Total
Group1	Group 1	10	7	21	1	39
	Group 2	6	3	10	1	20
	Group 3	16	10	34	1	61
Total		32	20	65	3	120

Chi-Square Tests

Value

df

Monte Carlo Sig. (2-sided)

Monte Carlo Sig. (1-sided)

			Asymptotic Significance (2- sided)	Significance	99% Confidence Interval		Significance	99% Confidence Interval	
					Lower Bound	Upper Bound		Lower Bound	Upper Bound
Pearson Chi-Square	.952 ^a	6	.987	.998 ^b	.997	.999			
Likelihood Ratio	.862	6	.990	.999 ^b	.998	1.000			
Fisher's Exact Test	1.696			.976 ^b	.972	.980			
Linear-by-Linear Association	.000 ^c	1	.982	1.000 ^b	1.000	1.000	.501 ^b	.488	.514
N of Valid Cases	120								

a. 4 cells (33.3%) have expected count less than 5. The minimum expected count is .50.

b. Based on 10000 sampled tables with starting seed 79996689.

c. The standardized statistic is -.022.

The Chi-square of these Monte Carlo samples showed at 95% confidence level that there is no significant correlation between the agreement degree on if it was easy to disclose carrier status result to the spouse, or it will be easy to disclose result to you're a future spouse and the marital status.

c) **It was easy for you to encourage your current spouse, or it will be easy for you to encourage your future spouse to undergo PMT**

Group1 * Agreement1 Crosstabulation

Count

		Agreement1					
		Agree	Disagree	Neutral	Strongly Agree	Strongly disagree	Total
Group1	Group 1	7	0	5	27	0	39
	Group 2	1	1	2	15	1	20
	Group 3	9	1	6	44	1	61
Total		17	2	13	86	2	120

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided) 99% Confidence Interval		Monte Carlo Sig. (1-sided) 99% Confidence Interval		Significance	Lower Bound	Upper Bound
				Significance	Lower Bound	Upper Bound	Significance			
Pearson Chi-Square	5.838 ^a	8	.665	.696 ^b	.684	.708				
Likelihood Ratio	6.560	8	.585	.729 ^b	.718	.741				
Fisher's Exact Test	5.968			.644 ^b	.631	.656				
Linear-by-Linear Association	.149 ^c	1	.699	.711 ^b	.700	.723	.361 ^b	.348		.373
N of Valid Cases	120									

a. 9 cells (60.0%) have expected count less than 5. The minimum expected count is .33.

b. Based on 10000 sampled tables with starting seed 846668601.

c. The standardized statistic is .386.

The Chi-square of these Monte Carlo samples showed at 95% confidence level that there is no significant correlation between the agreement degree on if it was easy for them to encourage your current spouse, or it will be easy for you to encourage your future spouse to undergo PMT the marital status.

D) Your carrier status effects your chances and choices for marriage

Group1 * Agreement1 Crosstabulation

Count

		Agreement1					Total
		Agree	Disagree	Neutral	Strongly Agree	Strongly disagree	
Group1	Group 1	12	2	10	13	2	39
	Group 2	6	5	4	4	1	20
	Group 3	12	14	22	11	2	61
Total		30	21	36	28	5	120

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
				Significance	99% Confidence Interval		Significance	99% Confidence Interval	
Pearson Chi-Square	10.900 ^a	8	.207	.207 ^b	.197	.218			
Likelihood Ratio	11.994	8	.151	.196 ^b	.186	.206			
Fisher's Exact Test	11.837			.135 ^b	.126	.144			
Linear-by-Linear Association	.267 ^c	1	.605	.624 ^b	.612	.637	.323 ^b	.311	.335
N of Valid Cases	120								

a. 5 cells (33.3%) have expected count less than 5. The minimum expected count is .83.

b. Based on 10000 sampled tables with starting seed 263739791.

c. The standardized statistic is -.517.

The Chi-square of these Monte Carlo samples showed at 95% confidence level that there is no significant correlation between the agreement degree that the carrier status affects the chances and choices for marriage and the marital status.

E) Your carrier status affects previous marriage plan (if any) in terms of cancellation, or you will cancel any marriage

Group1 * Agreement1 Crosstabulation

Count

		Agreement1					Total
		Agree	Disagree	Neutral	Strongly Agree	Strongly disagree	
Group1	Group 1	10	5	7	15	2	39
	Group 2	2	4	5	3	6	20
	Group 3	5	22	13	11	10	61
Total		17	31	25	29	18	120

Chi-Square Tests

				Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
			Asymptotic		99% Confidence			99% Confidence	
			Significanc	Significan	Lower	Upper	Significan	Lower	Upper
	Value	df	e (2-sided)	ce	Bound	Bound	ce	Bound	Bound
Pearson Chi-Square	21.581 ^a	8	.006	.004 ^b	.003	.006			

Likelihood Ratio	21.395	8	.006	.009 ^b	.007	.012			
Fisher's Exact Test	20.372			.006 ^b	.004	.008			
Linear-by-Linear Association	.155 ^c	1	.694	.729 ^b	.718	.741	.368 ^b	.355	.380
N of Valid Cases	120								

a. 4 cells (26.7%) have expected count less than 5. The minimum expected count is 2.83.

b. Based on 10000 sampled tables with starting seed 1831435319.

c. The standardized statistic is .394.

The Chi-square of these Monte Carlo samples showed at 95% confidence level that there is a significant correlation between the degree of agreement that their carrier status affects previous marriage plan (if any) in terms of cancelation, or you will cancel any marriage and the marital status.

Symmetric Measures

		Value	Asymptotic Standard Error ^a	Approximate T ^b	Approximate Significance	Monte Carlo Significance	99% Confidence Interval	
							Lower Bound	Upper Bound
Nominal by Nominal	Phi	.424			.006	.004 ^c	.003	.006
	Cramer's V	.300			.006	.004 ^c	.003	.006
	Contingency Coefficient	.390			.006	.004 ^c	.003	.006
Interval by Interval	Pearson's R	.036	.090	.392	.696 ^d	.729 ^c	.718	.741

Ordinal by Ordinal	Spearman Correlation	.017	.093	.187	.852 ^d	.855 ^c	.846	.864
N of Valid Cases		120						

- a. Not assuming the null hypothesis.
- b. Using the asymptotic standard error assuming the null hypothesis.
- c. Based on 10000 sampled tables with starting seed 1831435319.
- d. Based on normal approximation.

According to the Phi and Cramer's V correlation measurements there is a moderate correlation between the degree of agreement that their **carrier** status affects previous marriage plan (if any) in terms of cancelation, or you will cancel any marriage and the marital status.

- **For non carriers:** I would like to know the statistical significance in the answers across all the Five statements in table 3.

A) It was easy for you to disclose your carrier status result to the family

Group1 * Agreement1 Crosstabulation

Count

		Agreement1			Total
		Agree	Neutral	Strongly Agree	
Group1	Group 1	12	4	19	35
	Group 2	8	1	11	20
	Group 3	31	14	46	91
Total		51	19	76	146

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
				Significance	99% Confidence Interval		Significance	99% Confidence Interval	
					Lower Bound	Upper Bound		Lower Bound	Upper Bound
Pearson Chi-Square	1.725 ^a	4	.786	.797 ^b	.786	.807			
Likelihood Ratio	2.004	4	.735	.753 ^b	.742	.764			
Fisher's Exact Test	1.500			.844 ^b	.835	.854			
Linear-by-Linear Association	.005 ^c	1	.942	.945 ^b	.939	.951	.485 ^b	.472	.497
N of Valid Cases	146								

a. 2 cells (22.2%) have expected count less than 5. The minimum expected count is 2.60.

b. Based on 10000 sampled tables with starting seed 2129180967.

c. The standardized statistic is -.073.

The Chi-square of these Monte Carlo samples showed at 95% confidence level that there is no significant correlation between the degree of agreement that It was easy for you to disclose your carrier status result to the family and the marital status.

B) It was easy to disclose carrier status result to the spouse, or it will be easy to disclose result to you're a future spouse use

Group1 * Agreement1 Crosstabulation

Count

Agreement1

Total

		Agree	Disagree	Neutral	Strongly Agree	Strongly disagree	
Group1	Group 1	12	1	3	19	0	35
	Group 2	6	0	0	13	1	20
	Group 3	25	1	7	58	0	91
Total		43	2	10	90	1	146

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
				Significance	99% Confidence Interval		Significance	99% Confidence Interval	
					Lower Bound	Upper Bound		Lower Bound	Upper Bound
Pearson Chi-Square	9.613 ^a	8	.293	.303 ^b	.291	.315			
Likelihood Ratio	8.786	8	.361	.402 ^b	.389	.415			
Fisher's Exact Test	7.576			.450 ^b	.437	.463			
Linear-by-Linear Association	.681 ^c	1	.409	.433 ^b	.420	.446	.220 ^b	.209	.230 ^b
N of Valid Cases	146								

a. 8 cells (53.3%) have expected count less than 5. The minimum expected count is .14.

b. Based on 10000 sampled tables with starting seed 1156607048.

c. The standardized statistic is .825.

The Chi-square of these Monte Carlo samples showed at 95% confidence level that there is no significant correlation between the degree of agreement that It was easy to disclose carrier status result to the spouse, or it will be easy to disclose result to you're a future spouse use and the marital status.

C) It was easy for you to encourage your current spouse, or you it will be easy for you to encourage your future spouse to undergo PMT

Group1 * Agreement1 Crosstabulation

Count

		Agreement1					Total
		Agree	Disagree	Neutral	Strongly Agree	Strongly disagree	
Group1	Group 1	12	1	4	18	0	35
	Group 2	2	2	2	14	0	20
	Group 3	25	1	10	54	1	91
Total		39	4	16	86	1	146

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
				Significanc e	Lower Bound	Upper Bound	Significanc e	Lower Bound	Upper Bound
Pearson Chi-Square	8.988 ^a	8	.343	.353 ^b	.340	.365			
Likelihood Ratio	8.613	8	.376	.417 ^b	.404	.430			
Fisher's Exact Test	9.145			.277 ^b	.265	.288			
Linear-by-Linear Association	.522 ^c	1	.470	.474 ^b	.461	.487	.241 ^b	.230	.252

N of Valid Cases	146							
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- a. 8 cells (53.3%) have expected count less than 5. The minimum expected count is .14.
- b. Based on 10000 sampled tables with starting seed 2096426169.
- c. The standardized statistic is .723.

The Chi-square of these Mone Carlo samples showed at 95% confidence level that there is no significant correlation between the degree of agreement that It was easy for you to encourage your current spouse, or you it will be easy for you to encourage your future spouse to undergo PMT and the marital status.

D) Your carrier status effects your chances and choices for marriage

Group1 * Agreement1 Crosstabulation

Count

		Agreement1					Total
		Agree	Disagree	Neutral	Strongly Agree	Strongly disagree	
Group1	Group 1	10	5	12	6	2	35
	Group 2	2	3	3	11	1	20
	Group 3	24	13	20	28	6	91
Total		36	21	35	45	9	146

Chi-Square Tests

Value	df	Asymptotic Significance (2-sided)	Monte Carlo Sig. (2-sided)			Monte Carlo Sig. (1-sided)		
			Significance	99% Confidence Interval		Significance	99% Confidence Interval	
				Lower Bound	Upper Bound		Lower Bound	Upper Bound

Pearson Chi-Square	10.451 ^a	8	.235	.238 ^b	.227	.249			
Likelihood Ratio	10.562	8	.228	.282 ^b	.271	.294			
Fisher's Exact Test	10.031			.237 ^b	.226	.247			
Linear-by-Linear Association	.175 ^c	1	.675	.674 ^b	.662	.686	.355 ^b	.343	.368
N of Valid Cases	146								

a. 5 cells (33.3%) have expected count less than 5. The minimum expected count is 1.23.

b. Based on 10000 sampled tables with starting seed 1201225993.

c. The standardized statistic is .419.

The Chi-square of these Monte Carlo samples showed at 95% confidence level that there is no significant correlation between the degree of agreement that It was easy for you to encourage your current spouse, or you it will be easy for you to encourage your future spouse to undergo PMT and the marital status.

Knowledge	Group 1 (Single Engaged) N= 126	Group 2 (Married/ Divorced/Widowed with no children) N=52	Group 3: (Married/ Divorced/Widowed with healthy children) N= 230	Group 4: (Married/ Divorced/Widowed with at least one affected child) N= 251			
Q.1. Are you aware of the existence of genetic diagnosis in your family? At which stage did you have this information?				Before having your first affected child, Are you aware about the existence of the genetic diagnosis in your family?			
Aware	91% (n=115)	87% (n= 45)	Before Marriage: 69 % (n= 31)	79% (n=182)	Before Marriage 50.3% (n=92)	10% (n=26)	Before Marriage 42% (n=11)
			After Marriage: 31% (n=14)		After Marriage: 49.7% (n=90)		After Marriage: 58% (n=15)
Not Aware	9% (n=11)	13 % (n= 7)		21% (n=48)		90% (n=225)	
Q.2.Are you aware of the existence of premarital genetic testing for diseases diagnosed in your family? And, at which stage, did you have this information?							
Aware	88% (n=111)	80%(n=42)	Before Marriage: 61% (n=34)	62% (n=143)	Before Marriage 63% (n=90)	25% (n=63)	
			After Marriage: 19% (n=8)		After Marriage 37% (n=53)		
Not Aware	12% (n=15)	20% (n= 10)		38% (n=87)		75% (n= 188)	
Q.3.What was the source of your knowledge about PMT?							
Relatives	67.6% (n=75)	72% (n=30)		66% (n=93)		14% (n= 9)	
Health care	24.3% (n=27)	28% (n=12)		24% (n=34)		75% (n=47)	
Friends	4.5% (n=5)	0		5% (n=8)		8% (n=5)	
Others	3.6% (n=4)	0		5% (n=8)		3% (n=2)	
Q.4.“Genetic disease places a psychological burden on families” choses the option that describes your believe a bout the statement							

	For this group the question only appeared to those who are aware of the existence of PMT (n=111)		For this group this question appeared only for 142p due technical error. (N=142)	
Strongly Agree	65.8% (n=73)	83 % (n=43)	75% (n=107)	
Agree	21.6% (n=24)	15% (n=8)	13.4% (n=19)	
Neutral	10.8% (n=12)	2% (n=1)	9.2% (n=13)	
Disagree	1.8% (n=2)	0	(n=0)	
Strongly disagree	0	0	2.1% (n=3)	

Attitude and Practice	Group 1 (Single Engaged) n= 126	Group 2 (Married/ Divorced/Widowed with no children) n=52	Group 3: (Married/ Divorced/Widowed with healthy children) n= 230	Group 4: (Married/ Divorced/Widowed with at least one affected child) n= 251
				Did you disclose the diagnosis to other family members? N=225
				Yes 79% (n=177)
				No 21% (n=48)
“Prenatal genetic testing could act as a preventative option in the families with rare genetic disease” chooses the option that describes your believe a bout the statement				
Strongly Agree	82.5% (n=104)	90% (n=47)	89% (n=205)	85.6% (n=215)
Agree	12.7% (n=16)	10% (n=5)	10% (n=22)	9.5% (n=24)
Neutral	3.2% (n=4)	n=0	1 % (n=3)	3.5% (n=9)
Disagree	0.8% (n=1)	n=0	(n=0)	0.8% (n=2)
Strongly Disagree	0.8% (n=1)	n=0	(n=0)	0.4% (n=1)
				Did you advice other family members to conduct PMT before marriage?
				Yes 78% (n=196)
				No 22% (n=55)
Did you conduct the prenatal genetic testing for the disease diagnosed in your family before your marriage? If not are you willing to conduct it in the future? (For group1)				
Yes	59% (n=74)	77% (n= 40)	66% (n=152)	
NO	41% (n= 52) 96% (N=50) of them plan to take PMT in the future. 4% (n=2) did not conduct the test	23% (n= 12)	34% (n=78)	

	and not plan to do it in the future			
What was your carrier status?				
Carrier	53% (n=39)	50% (n=20)	40% (n=61)	
Not Carrier	47% (n=35)	50% (n=20)	60% (n=91)	
What was your main reasons for conducting the premarital genetic testing? (You can choose more than one option)				
A: To know my carrier status.	n= 59	n= 26	n=104	
B: Requested by my spouse and/or Family	n=13	n=10	n= 30	
C: To prevent transmission of the disease to my offspring	n= 53	n=26	n= 104	
D: I do not want to face the difficulties that my relative/s faced with the affected person/child	n=43	n=22	n=83	
E. Others	n=1 “One individual stated the reason as fearing to get the genetic disease symptoms”	n=2 -One underwent testing because he had symptoms, some of the autosomal recessive conditions associated with adult onset. Her testing is not considered as premarital testing, but it is a predictive testing. -Other one did the test based on his friend	n=1 One individual stated that my kid is brain paralyzed and I wanted to check if it's genetic related	

		advice		
Why do you disagree to undergo genetic testing (You can choose more than one option)				
A: I did not know about the genetic disease in the family.	4% (n=2)	n=6	n=50	
B: I am not interested to know my carrier status		n=0	n=6	
C. Afraid that test result will be in favour of my marital choice.		n=2	n=3	
D. I don't want to interfere to the god will.		n=4	n=0	
E. Fear of rejections by families That I want to marry from.		n=0	n=1	
C. Others		n=2 No risk 1. Parents did test and were not carriers. 2. Spouse did the test, and he/she is not carrier	n=12 Genetic testing not popular /known for them	
“The history of the genetic disease in your family affected your chances and choices for marriage” select the option that describes your believe a bout the statement choices for the marriage.				
Strongly Agree	21% (n= 26)	(29%) n=15	17% (n=40)	
Agree	30% (n= 38)	(21%) n=11	24% (n=55)	
Neutral	26% (n=33)	(31%) n=16	26% (n=60)	
Disagree	19% (n=24)	(14%) n=7	25% (n=57)	
Strongly disagree	4% (n= 5)	(5%) n=3	8% (n=18)	
What do you think is the appropriate timing for premarital genetic testing?				
At age 18 years	35.6% (n=45)	50% (n=26)	50% (n=115)	35.8% (n=90)
Before initiating the	57.9% (n=73)	50% (n=26)	44% (n=101)	57% (n=143)

marriage plan.				
After marriage	0.9% (n=1)	0	0	0.8% (n=2)
I don't know when the appropriate time for testing is.	5.6 % (n= 7)	0	6% (n=14)	6.4% (n=16)
What your advice to at-risk couple who are both carriers and at risk to have an affected child				
A: To continue with the engagement and marriage and keep faith in God.	2% (n=3)	6% (n=3)	0.4% (n=1)	1.2% (n=3)
B: Discussion will depend on the disease type and the severity of the symptoms.	12% (n=15)	8% (n=4)	9% (n=21)	8% (n=20)
C: To cancel / discontinue the engagement.	23% (n=29)	25% (n=13)	29.5% (n=68)	27.5% (n=69)
D: To proceed with the marriage if there is an emotional attachment between them.	1% (n=1)	2% (n=1)	0.9%(n=2)	0.8% (n=2)
E: To proceed with the marriage if there is family pressure	n=0	2% (n=1)	0	0
F: To consult a genetic counsellor/geneticist	61% (n= 77)	57% (n=30)	59% (n=136)	62.5% (n=157)
G: Others: here statement written by one individual	1% (n=1) One individual stated: "many people undergo premarital testing, but the couple did not listen to any advice".		0.9% (n=2) No statement was written.	0
To which extend you agree with this statement "The premarital genetic testing for rare genetic disorders should be voluntary not mandatory"				
chooses the option that describes your believe a bout the statement				
Strongly Agree	10% (n =13)	6% (n=3)	12% (n=27)	17.5% (n=44)
Agree	12% (n=15)	15% (n=8)	13% (n=30)	16.3% (n=41)
Neutral	15% (n=19)	19% (n=10)	15% (n=34)	11.6% (n=29)
Disagree	32% (n=40)	25% (n=13)	28%(n=65)	24.3% (n=61)
Strongly disagree	31% (n=39)	35%(n=18)	32% (n=74)	30.3% (n=76)

Likert scale Utilized in the comparison between carriers and Non carriers

Option	Mean Interval	Decision/ Degree of approval
Strongly Disagree	1.00 – 1.80	Very Low
Disagree	1.81 – 2.60	Low
Neutral	2.61 – 3.40	Medium
Agree	3.41 – 4.20	High
Strongly Agree	4.21 – 5.00	Very High

Table 3: attitude and practice of carriers across groups 1,2 and 3

	Group 1 (Single / Engaged)	Group2 (Married with no children)	Group3 (Married with healthy Chlidren)
Number of carriers each group (Total=120)	N=39	N=20	N= 61
A. It was easy to disclose carrier status result to the family			
Strongly disagree	2.6%	0	0%
Disagree	5.1%	15%	4.9%
Neutral	10.3%	15%	18%
Agree	25.6%	25%	32.8%
Strongly Agree	56.4%	45%	44.3%
Mean	4.28	4.00	4.16
Std	1.025	1.12	0.90
Decision	Very high	high	High
B. It was easy to disclose carrier status result to the spouse, or it will be easy to disclose result to you're a future spouse			
Strongly disagree	2.6%	5%	1.7%
Disagree	0%	0%	0%
Neutral	17.9%	15%	16.4%
Agree	25.6%	30%	26.2%
Strongly Agree	53.8%	50%	55.7%
Mean	4.28	4.20	4.34
Std	0.944	1.06	0.87
Decision	Very high	High	Very high
C. It was easy for you to encourage your current spouse, or it will be easy for you to encourage your future spouse to undergo PMT			

Strongly disagree	0%	5%	1.6%
Disagree	0%	5%	1.6%
Neutral	12.8%	10%	9.8%
Agree	17.9%	5%	14.8%
Strongly Agree	69.2%	75%	72.2%
Mean	4.56	4.40	4.54
Std	0.718	1.19	0.87
Decision	Very high	Very high	Very high
D. Your carrier status effects your chances and choices for marriage			
Strongly disagree	5.1%	5%	3.3%
Disagree	5.1%	25%	23%
Neutral	25.6%	20%	36.1%
Agree	30.8%	30%	19.7%
Strongly Agree	33.3%	20%	18%
Mean	3.82	3.35	3.26
Std	1.121	1.22	1.11
Decision	High	Medium	Medium
E. Your carrier status affects previous marriage plan (if any) in terms of cancelation, or you will cancel any marriage			
Strongly disagree	5.1%	30%	16.4%
Disagree	12.8%	20%	36.1%
Neutral	17.9%	25%	21.3%
Agree	25.6%	10%	8.2%
Strongly Agree	38.5%	15%	18%
Mean	3.79	2.60	2.75
Std	1.239	1.43	1.33
Decision	High	Low	Medium
Overall Factor for all statements combined			
Mean	4.15	3.71	3.81
Std	0.51	0.59	0.59
Decision	High	High	High

Table 4: attitude and practice of non carriers carriers across groups 1,2 and 3

	Group 1 (Single/ Engaged)	Group2 (Married with no children)	Group3 (Married with healthy Children)
Number of non- carriers for each group (Total =146)	n = 35	n =20	n= 91

A. It was easy for you to disclose your carrier status result to the family			
Strongly disagree	0%	0%	0%
Disagree	0%	0%	0%
Neutral	11.4%	5%	15.4%
Agree	34.3%	40%	34.1%
Strongly Agree	54.3%	55%	50.5%
Mean	4.43	4.50	4.35
Std	0.698	0.61	0.74
Decision	Very high	Very high	Very high
B. It was easy to disclose carrier status result to the spouse, or it will be easy to disclose result to you're a future spouse use			
Strongly disagree	0%	5%	0%
Disagree	2.9%	0%	1.1%
Neutral	8.6%	0%	7.7%
Agree	34.3%	30%	27.5%
Strongly Agree	54.3%	65%	63.7%
Mean	4.40	4.50	4.54
Std	0.775	0.95	0.69
Decision	Very high	Very high	Very high
C. It was easy for you to encourage your current spouse, or you it will be easy for you to encourage your future spouse to undergo PMT			
Strongly disagree	0%	0%	1.1%
Disagree	2.9%	10%	1.1%
Neutral	11.4%	10%	11%
Agree	34.3%	10%	27.5%
Strongly Agree	51.4%	70%	59.3%
Mean	4.34	4.40	4.42
Std	0.802	1.05	0.82
Decision	Very high	Very high	Very high
D. Your carrier status effects your chances and choices for marriage			
Strongly disagree	5.7%	5%	6.5%
Disagree	14.3%	15%	14.3%
Neutral	34.3%	15%	22%
Agree	28.6%	10%	26.4%
Strongly Agree	17.1%	55%	30.8%
Mean	3.37	3.95	3.60
Std	1.114	1.36	1.25
Decision	High	High	High
Overall Factor for all statements combined			
Mean	4.14	4.34	4.23
Std	0.58	0.59	0.55

Decision	High	Very high	Very high